

2024

Bachelor in Business Administration

Course Title: Financial Derivatives

Code No: **BNK202** Semester: **8th Semester**

Full Marks: 60 Pass Marks: 24 Time: to expiration.4.When pricing a put with the binomial model, the up and down probabilities are reversed.5.Both put option buyer and seller have the potential for unlimited losses.6.If the initial margin is Rs 5,000, the maintenance margin is Rs 3,500 and your margin balance is Rs 4,000, you will receive margin call of Rs 500.7.A future contract can have negative value.8.A swap involving two floating rates in interest rate swap is called a basis swap.9.In case of futures, the initial margin is paid only by the seller and not the buyer.10.Three month put option on stock of XYZ company is currently selling for Rs 20. The exercise price of put is Rs 250 and the current market price of stock is Rs 235. The time value of put is Rs 5. Section "B" Short Answer Questions:[6 × 5 = 30]11.Define derivative markets. Describe the major functions of derivative markets in an economy.12.Suppose you believe that the price of a particular underlying stock, currently selling at Rs 140, will decrease considerably in the next six months. You decide to purchase a put option expiring in six months on this underlying stock. The put option has an exercise price of Rs 130 and sells for Rs 12.a. Determine the gain or loss for you if the possible ending prices of the underlying stock six months are Rs 150, Rs 130, Rs 120 and Rs 110.b. Determine the breakeven price of the underlying at expiration. Check that your answer is consistent with the solution to Part a of this problem.c. What is the maximum profit and loss that you can have?13.The call option of a certain company has an exercise price of Rs 250 and a maturity date 6 months from now. The stock price is Rs 260. You have made a careful study of the stock's volatility and concluded that a standard deviation of 0.30 is appropriate for the next 6 months. Currently, the annual rate on short-term treasury bills is 6 percent. What is the value of call option?14.The current price of the underlying asset is Rs 900. The maturity period of a futures contract on this underlying asset is 30 days. The annual risk-free interest rate is 5 percent.a. Calculate the futures price.b. Determine the futures price if the underlying asset's storage costs is Rs 30 at expiration.c. Describe the similarities of futures and forward contract.15.You enter into a long futures position in 1 contract in gold at a futures price of Rs 8,400,000 per kg. You purchased one kg gold futures. The contract size is 1 kg. The broker requires Rs 200,000 initial margin deposit per contract and a maintenance margin is 150,000 per contract.Day12345Settlement price (Rs)8,380,0008,304,0008,300,0008,650,0008,350,000Calculate the daily gain or loss, cumulative gain or loss, margin balance and margin call if any. Determine the price level that would trigger a margin call. If investor does not deposit margin call amount, what will happen?16.An asset manager wishes to reduce her exposure to small-cap stocks and increase her exposure to fixed-income securities. She seeks to do so using an equity swap. She agrees to pay a dealer the return on a small-cap index and the dealer agrees to pay the manager a fixed rate of 5.5 percent. For each of the scenarios listed below, calculate the overall payment six months later and indicate which party makes the payment. Assume that payments are made semiannually (180 days per period) and there are 360 days in each year. The notional principal is Rs 50,000,000.a. The value of the small-cap index starts off at 234.10 and six months later is at 238.41.b. The value of the small-cap index starts off at 234.10 and six months later is at 241.27. Section "C" Comprehensive answer questions:[2 × 10 = 20]17.Consider the binomial option pricing model for an American call option. This call has two periods in a year before expiration. Each binomial has the period of six months. Current stock price is Rs 400 per share and exercise price is Rs 350. The risk-free rate is 10 percent (5 percent per period). At the end of each binomial period, the stock price either increases or decreases by 20 percent per period. What is the maximum price you should be willing to pay for this American call option? What would be your investment strategy if the market price of this call is Rs 30?18.Consider the common stock of ABC company. The current price of stock is Rs 170 per share. Call and put option are available on this stock. Call option on ABC stock with Rs 170 exercise price and three months to expiration are currently selling at Rs 20 and put options on this stock with Rs 170 exercise price and three months to expiration are selling for Rs 10. Possible stock prices at the expiration dates are Rs 140, Rs 150, Rs 160, Rs 170, Rs 180, Rs 190, and Rs 200.a. Buy one call contract and hold it until the options expire. Determine the gains or losses, breakeven price, maximum gain and loss of this transaction.b. Buy one put contract and hold it until the options expire. Determine the gains or losses, breakeven stock price, maximum gain and loss on this transaction.c. Buy 100 shares of stock and write one call contract and hold the position until expiration. Determine gain or loss, breakeven stock price, the maximum profit, and the maximum loss.

Candidates are required to answer the questions in their own words as far as practicable.

Section A

Brief Answer Questions:[10 × 1 = 10]

1. Value of a financial derivative depends upon value of underlying asset.
2. In the money put option has positive intrinsic value.
3. There is direct / positive relationship between value of call option with time to expiration.
4. When pricing a put with the binomial model, the up and down probabilities are reversed.
5. Both put option buyer and seller have the potential for unlimited losses.
6. If the initial margin is Rs 5,000, the maintenance margin is Rs 3,500 and your margin balance is Rs 4,000, you will receive margin call of Rs 500.
7. A future contract can have negative value.
8. A swap involving two floating rates in interest rate swap is called a basis swap.
9. In case of futures, the initial margin is paid only by the seller and not the buyer.
10. Three month put option on stock of XYZ company is currently selling for Rs 20. The exercise price of put is Rs 250 and the current market price of stock is Rs 235. The time value of put is Rs 5.

Section B

Short Answer Questions:[6 × 5 = 30]

11. Define derivative markets. Describe the major functions of derivative markets in an economy.
12. Suppose you believe that the price of a particular underlying stock, currently selling at Rs 140, will decrease considerably in the next six months. You decide to purchase a put option expiring in six months on this underlying stock. The put option has an exercise price of Rs 130 and sells for Rs 12.
 - a. Determine the gain or loss for you if the possible ending prices of the underlying stock six months are Rs 150, Rs 130, Rs 120 and Rs 110.
 - b. Determine the breakeven price of the underlying at expiration. Check that your answer is consistent with the solution to Part a of this problem.
 - c. What is the maximum profit and loss that you can have?
13. The call option of a certain company has an exercise price of Rs 250 and a maturity date 6 months from now. The stock price is Rs 260. You have made a careful study of the stock's volatility and concluded that a standard deviation of 0.30 is appropriate for the next 6 months. Currently, the annual rate on short-term treasury bills is 6 percent. What is the value of call option?
14. The current price of the underlying asset is Rs 900. The maturity period of a futures contract on this underlying asset is 30 days. The annual risk-free interest rate is 5 percent.
 - a. Calculate the futures price.
 - b. Determine the futures price if the underlying asset's storage costs is Rs 30 at expiration.
 - c. Describe the similarities of futures and forward contract.
15. You enter into a long futures position in 1 contract in gold at a futures price of Rs 8,400,000 per kg. You purchased one kg gold futures. The contract size is 1 kg. The broker requires Rs 200,000 initial margin deposit per contract and a maintenance margin is 150,000 per contract.

Day12345Settlement price

(Rs)8,380,0008,304,0008,300,0008,650,0008,350,000Calculate the daily gain or loss, cumulative gain or loss, margin balance and margin call if any. Determine the price level that would trigger a margin call. If investor does not deposit margin call amount, what will happen?

- 16.** An asset manager wishes to reduce her exposure to small-cap stocks and increase her exposure to fixed-income securities. She seeks to do so using an equity swap. She agrees to pay a dealer the return on a small-cap index and the dealer agrees to pay the manager a fixed rate of 5.5 percent. For each of the scenarios listed below, calculate the overall payment six months later and indicate which party makes the payment. Assume that payments are made semiannually (180 days per period) and there are 360 days in each year. The notional principal is Rs 50,000,000.a. The value of the small-cap index starts off at 234.10 and six months later is at 238.41.b. The value of the small-cap index starts off at 234.10 and six months later is at 241.27.
- 17.** Consider the binomial option pricing model for an American call option. This call has two periods in a year before expiration. Each binomial has the period of six months. Current stock price is Rs 400 per share and exercise price is Rs 350. The risk-free rate is 10 percent (5 percent per period). At the end of each binomial period, the stock price either increases or decreases by 20 percent per period. What is the maximum price you should be willing to pay for this American call option? What would be your investment strategy if the market price of this call is Rs 30?
- 18.** Consider the common stock of ABC company. The current price of stock is Rs 170 per share. Call and put option are available on this stock. Call option on ABC stock with Rs 170 exercise price and three months to expiration are currently selling at Rs 20 and put options on this stock with Rs 170 exercise price and three months to expiration are selling for Rs 10. Possible stock prices at the expiration dates are Rs 140, Rs 150, Rs 160, Rs 170, Rs 180, Rs 190, and Rs 200.a. Buy one call contract and hold it until the options expire. Determine the gains or losses, breakeven price, maximum gain and loss of this transaction.b. Buy one put contract and hold it until the options expire. Determine the gains or losses, breakeven stock price, maximum gain and loss on this transaction.c. Buy 100 shares of stock and write one call contract and hold the position until expiration. Determine gain or loss, breakeven stock price, the maximum profit, and the maximum loss.